Danish Nadeem

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Portfolio Website: https://d-nadeem.github.io/portfolio/

Skills

Soft Skills: Project Management, Teamwork, Root Cause Analysis, Excellent Communication Skills, Adaptability
Technical Skills: SolidWorks, Onshape, Raspberry Pi, Mechanical & Electrical Systems Design, Sensors & Actuators, Automation, Prototyping & Testing, Technical Documentation & SOPs, Engineering Drawing (Mechanical & Electrical)

Lincoln Electric **2022 - 2023**
R&D Engineering Intern *East York, Toronto*

- **Collaborated** with engineering teams on **mechanical systems**, **electrical systems**, and applying **testing methodologies** to **prototypes** and **final products**.
- Conducted **field testing** and **performance evaluations** of prototype machinery, documenting findings used for **engineering reports** and review.
- Supported prototype servicing and performed **diagnostics**, developing hands-on **troubleshooting** and **maintenance** skills.
- Prepared **wiring diagrams**, **bill of materials**, **fabrication drawings** and developed instructions **for assembly** for production
- Contributed to **cost-saving** projects through **design revisions** and **material efficiency** improvements.
- Developed detailed **SOPs** that enabled team members to independently conduct test without needing prior knowledge or assistance.

Projects

Smart Airport Cart | **Capstone Project – Public Navigation System**

- **Led** a **team of four** to design and implement a marker-based **indoor navigation system** to guide elderly users through **airport terminals**.
- Developed a **touchscreen interface** for **user feedback** and **location instruction**, focusing on clarity and **accessibility**.
- Maintained the project within **budget constraints** with the use of **decision matrices**, soliciting **feedback from supervisor**, other teams, and experts, **cost reductions**.
- Presented to a **public audiences** during the **capstone exhibition**, highlighting strong communication and public-facing **technical explanation**.

Ball Balancing using a 3RRS Parallel Manipulator | **RT Embedded Systems Project**

- Achieved **project success** by leading the team to have the manipulator balance the ball in a **live demo**. Strategically **divided tasks** based on team members' skill sets and reallocated responsibilities to **overcome challenges**
- **Reduced costs** compared to the reference project by **optimizing hardware** and **program efficiency**
- Improved touchscreen program response time and **reduced computational load** by implementing direct port manipulation, moving average and median filters

2D and 3D Octopus Arm End Effector | *Onshape* **Personal Project**

- **Redesigned, octopus-inspired robotic arm**, **simplifying structure** and **reducing component count** to **cut manufacturing cost and assembly time**
- **TPU to PLA** conversion, **increasing compression strength, load capacity**, and **print reliability** on standard FDM printers
- Improved **serviceability** by developing a modular, maintenance-friendly design that **reduced repair time** and increased system robustness

Palletization Work Cell | **Robotics and Automation Project**

- **Collaborated** with a team to design and develop a **palletizing workcell** focusing on compliance with **CSA Safety Regulations** CSA Z434 and CSA Z432
- Documented and validated **safety protocols** through CAD assemblies and **risk assessments**
- Ensured **regulatory compliance** by reviewing and applying **CSA standards** for **industrial robotic manipulators**

Education

Ontario Tech University
Mechatronics Engineering B.Eng. *2025*